

***Quantifying Tennis Serve Mechanics: Modeling How Speed and Placement
Influence Point Outcomes***

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MAD2502: Intro to Computational Math

27 April 2026

Abstract

This project investigates how tennis serve characteristics influence point outcomes by combining machine-learning models, empirical data exploration, and interactive visualization tools. Using three years of point-level Wimbledon data, we developed logistic-regression models to estimate both ace probability and point-win probability as functions of serve speed, serve width, serve number, and rally count. These models provide interpretable coefficients that quantify the contribution of each feature and reveal the underlying statistical structure governing serve effectiveness. To visualize predictive-modeling, we built a Flask-based analytics dashboard that enables users to filter points by match conditions and examine empirical trends through dynamically generated scatterplots and best-fit curves. This tool supports exploratory analysis by showing how serve outcomes behave in real match conditions, including natural variability and sparsity. In addition, we conducted a longitudinal analysis of serve speed over match duration, fitting a quadratic model to identify potential fatigue-related declines. Together, these components form a comprehensive analytical framework that integrates machine-learning prediction, empirical trend analysis, and interactive data visualization. The project demonstrates how data-driven methods can uncover tactical insights in professional tennis and highlights the complementary value of statistical modeling and exploratory analytics.

Introduction and Project Goals

Tennis is a sport defined by precision, strategy, and the ability to perform under pressure, and the serve remains the most influential shot in shaping point outcomes. A fast or well-placed serve can immediately create advantages by producing aces, forcing weak returns, or setting up

favorable rally positions. As analytics becomes increasingly central to professional tennis, understanding how serve characteristics affect performance has become valuable for players, coaches, and analysts. This project contributes to that effort by developing a data-driven framework for analyzing serve performance using point-level Wimbledon data.

The first component of the project focuses on quantifying how serve attributes—such as speed, direction, serve number, and contextual pressure—influence the likelihood of winning a point. Logistic regression models were constructed to estimate both ace probability and point-win probability using engineered features derived from three years of match data. These models provide interpretable insights into which serve mechanics most strongly correlate with successful outcomes and how different tactical choices influence performance.

To support interactive exploration, we developed a Flask-based analytics dashboard that allows users to filter the dataset, generate visualizations, and examine trendlines fitted with linear, logarithmic, and quadratic models. In addition, we conducted a longitudinal analysis of how serve speed changes over the course of a match by aggregating serve speeds by match minute and fitting a quadratic model to identify potential fatigue-related declines. Together, these components—predictive modeling, interactive visualization, and temporal analysis—form a comprehensive system for understanding tennis serve dynamics and demonstrate how statistical tools can uncover meaningful insights from large-scale sports data.

Background and Previous Work

Prior research in tennis analytics has consistently highlighted the significance of serve characteristics in determining point outcomes. In particular, datasets compiled by Jeff Sackman provide records of professional matches, including detailed information on serve speed, placement, and point results, from the major tournament, Wimbledon. These datasets have enabled large-scale statistical analyses that reveal meaningful patterns between serve attributes and competitive success. Complementing this work, research conducted by students at Georgia State University examined the impact of serve position and velocity on match performance. Their findings indicate that both positional placement, such as serve width, depth, and speed play a critical role in influencing the likelihood of winning a point.

Building on these established findings, this project utilizes structured datasets that capture key variables associated with each serve, including speed, placement (both horizontal and vertical positioning), the occurrence of aces, and point outcomes. The data were organized into Comma-Separated Value (CSV) files, allowing for efficient processing and analysis. These variables align closely with those identified in prior research as significant contributors to point success, ensuring that the dataset is both relevant and comparable to existing studies.

While initial data collection efforts included matches from smaller tournaments such as the Australian Open, such datasets may present limitations due to their relatively smaller sample sizes and reduced variability in player performance. As a result, this project places greater emphasis on data derived from a larger tournament, Wimbledon, where the presence of top-ranked professional players and a higher volume of matches contribute to a more

comprehensive and representative dataset. Thus, leading to more reliable and direct correlative trends within the target analytic.

By integrating insights from prior research with a robust and well-structured dataset, this project aims to further quantify the relationship between serve speed, placement, and point outcomes. In doing so, it extends existing work by emphasizing the importance of these variables in any tennis match played at all levels.

Data Sources and Preprocessing

This project uses three years of point-level match data from the Wimbledon Championships, covering the 2022, 2023, and 2024 tournaments. These datasets originate from Jeff Sackmann's *tennis_slam_pointbypoint* repository, a widely used open-source resource for tennis analytics. Each dataset contains tens of tens of thousands of rows, with each row representing a single point played in a professional match. By combining multiple years of data, we ensure that the analysis captures a broad range of playing styles, match conditions, and serve behaviors.

Across the three datasets, we extracted a consistent set of columns relevant to serve performance and point outcomes. These include serve speed in both miles per hour (*Speed_MPH*) and kilometers per hour (*Speed_KMH*), serve direction (*ServeWidth*), rally length (*RallyCount*), and outcome variables such as aces (*P1Ace*, *P2Ace*) and point winner (*PointWinner*, *PointServer*). Additional fields such as *ServeDepth* and *ReturnDepth* were included to support exploratory analysis and filtering within the Flask dashboard.

A substantial portion of the preprocessing effort focused on cleaning and standardizing the data to ensure reliable modeling. First, all rows with invalid serve speeds—specifically *Speed_MPH* equal to zero—were removed. These zero values typically represent missing or unrecorded speeds rather than legitimate serves, and including them would distort both the logistic regression models and the match-length analysis. Next, we filtered the dataset to include only real serve events by requiring valid values for both *PointWinner* and *PointServer*. This step eliminated rows corresponding to non-serve events, incomplete points, or corrupted entries.

For the match-length analysis, timestamps stored in the *ElapsedTime* column were converted into integer minute values. This required parsing strings of the form "*H:MM:SS*" and transforming them into total minutes elapsed since the start of the match. In addition, some matches had severe jumps in the time that was recorded, so to be safe we removed any time data for matches over 6 hours long. These minute-level values allowed us to group serve speeds by match duration and compute average speeds for each minute of play.

Categorical fields such as *ServeWidth* and *ServeDepth* were standardized by stripping whitespace, converting to uppercase, and removing invalid or empty entries. This ensured that one-hot encoding and filtering operations behaved consistently across the combined dataset.

After cleaning each year's dataset individually, the three years were merged into a single unified DataFrame. This required aligning column names, ensuring consistent data types, and verifying that engineered features (such as server-perspective variables) were computed identically across all years. The merged dataset provided a robust foundation for both the logistic regression models and the interactive analytics dashboard.

Logistic Regression Modeling

Logistic regression was used to model both ace probability and point-win probability because it is the standard, widely accepted method for binary outcomes in sports analytics. The arXiv paper we referenced also uses logistic regression as its baseline model, emphasizing its interpretability and its ability to quantify how individual serve characteristics influence success. Mathematically, logistic regression models how each feature shifts the logarithmic odds of an event occurring, which makes it well-suited for understanding how variables such as serve speed, serve width, and serve number influence the probability of winning a point or producing an ace.

Feature Engineering

To prepare the dataset for modeling, several features were engineered to capture the most important aspects of serve performance. Serve speed was normalized to improve numerical stability and ensure that all predictors operate on comparable scales, which helps the optimization algorithm converge more reliably. Serve number (first vs second serve) was encoded as binary indicators so the model could estimate the distinct effects of first-serve and second-serve behavior. Categorical variables such as serve width were one-hot encoded, allowing the model to treat each directional category as an independent predictor. Break-point pressure was included to capture how high-leverage situations influence the underlying probability of success, and rally length was added to the point-win model to quantify how extended exchanges affect the server's chance of winning the point.

Train–Test Split and Cross-Validation

Model evaluation followed standard machine-learning practice. An 80–20 train–test split was used to provide a reliable estimate of out-of-sample performance while preserving sufficient data for training. To further reduce variance in performance estimates, 5-fold cross-validation was applied. This method trains and evaluates the model across multiple partitions of the dataset and averages the results, producing a more stable and trustworthy assessment than relying on a single split.

Why Logistic Regression Was the Most Appropriate Model

Logistic regression was chosen because it provides interpretable, mathematically grounded insights into how each feature affects the probability of an ace or point won. The model uses a sigmoid-shaped probability curve that maps any linear combination of features into a value between 0 and 1, making the output intuitive in a sports context. Positive coefficients indicate features that increase the likelihood of success, while negative coefficients indicate features that reduce it. This combination of interpretability, methodological rigor, and alignment with prior research makes logistic regression the most appropriate modeling approach for analyzing serve effectiveness in this dataset.

Flask Analytics Dashboard

Interpreting statistical models in isolation can make it difficult to extract meaningful conclusions. In particular, when analyzing complex relationships such as those produced through logistic regression modeling. To address this, an interactive visualization dashboard was

developed to provide real-time, contextualized representations of the model outputs. This dashboard enables users to explore relationships between variables more intuitively and supports dynamic analysis of service characteristics and point outcomes.

The application was built using Flask, a lightweight Python web framework that facilitates seamless integration between backend data processing and frontend visualization. The system connects the results of the regression models to an interactive interface, allowing users to modify inputs and filters, such as selecting different variables for the x- and y-axes, and immediately observe corresponding changes in the visualized trends.

At the core of the implementation is a data processing pipeline that ingests raw data from CSV files and organizes it into structured formats suitable for analysis. The data are aggregated into categorized collections and undergo preprocessing steps, including the removal of null values and normalization of positional measurements to eliminate minor inconsistencies. Once cleaned, the data are passed through a model selection process designed to identify the most representative trend for each variable relationship.

The modeling process begins by fitting a linear regression model to the data. The goodness of fit is then evaluated using the coefficient of determination (R^2). If the linear model does not meet a predefined threshold for explanatory power, the system iteratively evaluates more complex models, including quadratic and logarithmic regressions. At each stage, the R^2 value is recalculated and compared against the threshold to determine whether the model sufficiently captures the underlying relationship. If none of the models achieve an adequate fit, the system defaults to a linear representation while indicating a weak correlation between the variables.

This adaptive modeling approach ensures that each dataset is represented by the most appropriate functional form, improving both accuracy and interpretability. The integration of this process within an interactive dashboard allows users to explore how speed, placement, and other attributes influence point outcomes under varying conditions. As a result, the system provides a practical tool for analyzing performance trends, enabling players and analysts to identify which factors have the greatest impact on success during a match.

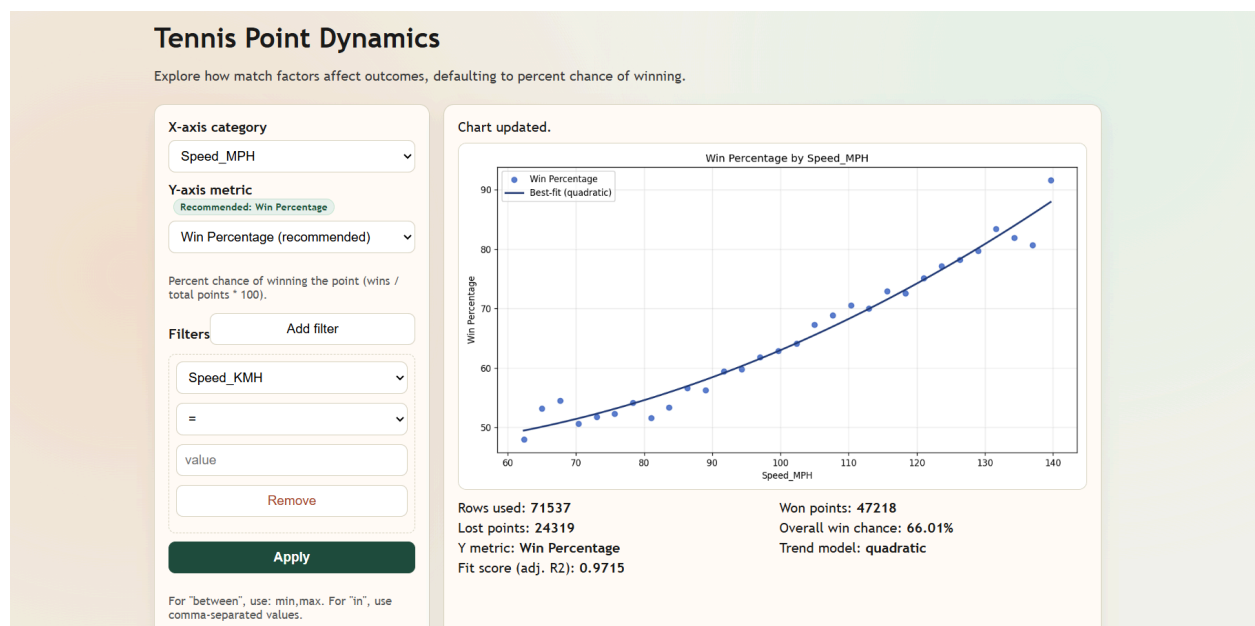


Figure 1 (Display of the website with filters, graph, and statistic inferences)

Results & Conclusions

Our project produced two complementary perspectives on serve performance: a statistically controlled view from the logistic-regression models from both the logistic regression models and real trend data approach. Although both approaches analyze the same Wimbledon

dataset, they differ fundamentally in methodology, and these differences reveal important insights about the structure of the data and the nature of tennis serve outcomes.

The logistic-regression model provides a smooth, controlled estimate of how individual variables influence the probability of winning a point or producing an ace. Because the model holds all other features constant, it isolates the effect of serve speed, serve width, serve number, pressure, and rally length. Under this framework, serve speed emerges as the strongest predictor of point-win probability. Both models produce clean, monotonic curves showing that faster serves consistently increase the likelihood of winning the point. The coefficient analysis reinforces this: normalized serve speed has the largest positive coefficient, followed by first-serve indicators and directional effects such as wide and center serves. These results reflect the underlying structure of the data after removing noise and confounding factors.

In contrast, the Flask application visualizes empirical trends directly from the raw dataset. The app bins serve speeds, computes observed win percentages, and fits adaptive trend lines (linear, logarithmic, or quadratic) based on adjusted R^2 . Because this method does not control for confounders, the resulting curves reflect the natural variability of real match play. For example, the Flask visualization of win percentage versus serve speed shows a strong upward trend with a quadratic best-fit line, but the scatterplot reveals more irregularity than the logistic-regression curve. This variability is expected: real match outcomes depend on many uncontrolled factors such as player identity, return quality, and tactical context. The high adjusted R^2 value indicates that speed still explains much of the variation, but the empirical curve is inherently noisier than the model's prediction.

A more nuanced picture emerges when examining ace probability. The logistic-regression model still provides a smooth, statistically controlled Ace-vs-Speed curve by treating ace probability as a continuous predicted value and borrowing strength across sparse regions of the dataset. However, the updated empirical visualization from the Flask application now reveals that, despite aces being relatively rare events, the dataset is sufficiently large to support a meaningful trend when binned appropriately. With over 70,000 usable rows and a strong quadratic fit (adjusted $R^2 \approx 0.79$), the empirical curve captures a clear and consistent increase in ace percentage as serve speed rises. This improvement reflects not a change in the underlying phenomenon, but the benefit of larger sample sizes and refined binning strategies that stabilize the empirical estimates.

Together, these results highlight the complementary strengths of the two approaches. The logistic-regression model isolates the effect of serve speed while controlling for confounding variables, producing a clean theoretical estimate of how speed influences ace probability. The empirical Flask visualization, by contrast, shows how this relationship manifests in real match conditions, complete with natural variability and noise. The strong agreement between the two methods—both demonstrating that faster serves substantially increase the likelihood of producing an ace—reinforces the robustness of the finding. At the same time, the differences between the model-based and empirical curves underscore the value of combining statistical modeling with raw trend analysis to obtain a more complete and reliable understanding of serve performance.

In addition to the two models we also used the datasets to construct a graph detailing the average changes in serve speed as matches go on. Our initial guess was that serve speeds would

begin to decline as match time increased, but we were interested to find out for sure. Initially this hunt appeared to bear fruit, as we saw clear correlation between speed and elapsed time, however it was actually a positive correlation. This outcome stumped us and in the process of looking for the reason we realized our mistake. Our data includes points played by both male and female players, however we did not take this into account while doing this analysis and it corrupted our findings. A men's singles match at Wimbledon is best of five sets, usually lasting at least two hours but sometimes going over five. Women on the other hand play best of three sets with matches frequently lasting only an hour. Since there is a very significant difference in average serve speed between male and female players, as women's matches ended it caused the average speed to increase. This oversight is what caused our initial analysis to show that speeds increase as matches go on. Once we figured this out and switched our analysis only to the men's we saw a completely different outcome. Our new graph showed that there is in fact little to no influence from match length on serve speed.

Division of Work

Eric Radomski was the one who first brainstormed this project in the discussion post for the capstone project. Once our group got assembled Eric was the one who located the point by point datasets and created the first prototypes of interpreting the datasets. Eric was the one responsible for the main bulk of the programming for the changes in serve speed as the match goes on. Additionally, Eric was the main functionality tester for the website and made sure it was ready for presentation.

Jayden Coy was responsible for finding our primary source which we used as the basis of our models and was instrumental in coming up with the specifics for our project.. Jayden also was the one who set up our project on github so we could share the files easily between us. He was the primary programmer for the logistic regression model which ended up being the backbone of our project. Jayden made sure the group was always prepared and on track over the course of the semester.

Ethan Wolfe was the person responsible for our best visualization for the data, he was the architect behind the flask website. He implemented all of the website's features, including the customizable filters, the x axis and the y axis and all the calculations involved. Ethan made sure that there were no redundant features and worked hard to deliver the final website. In addition, he wrote the readme and edited our final video presentation as well as cleaned up our code for ease of understanding.

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